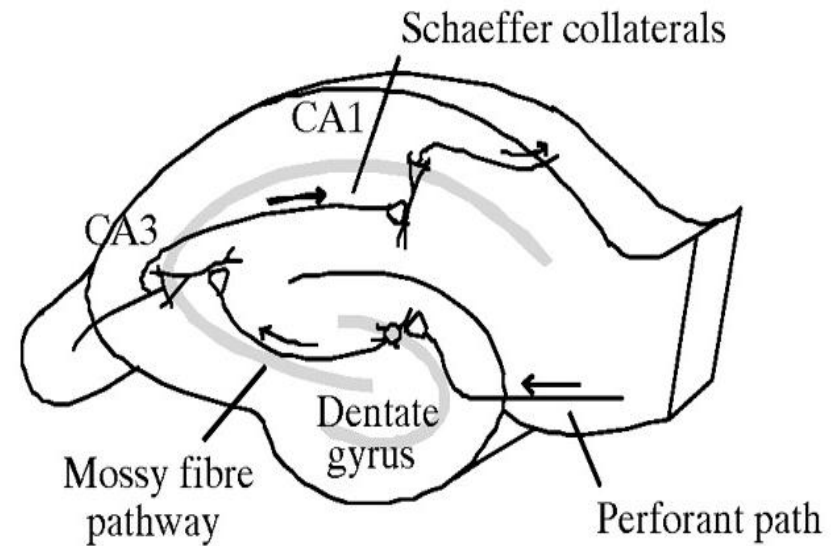
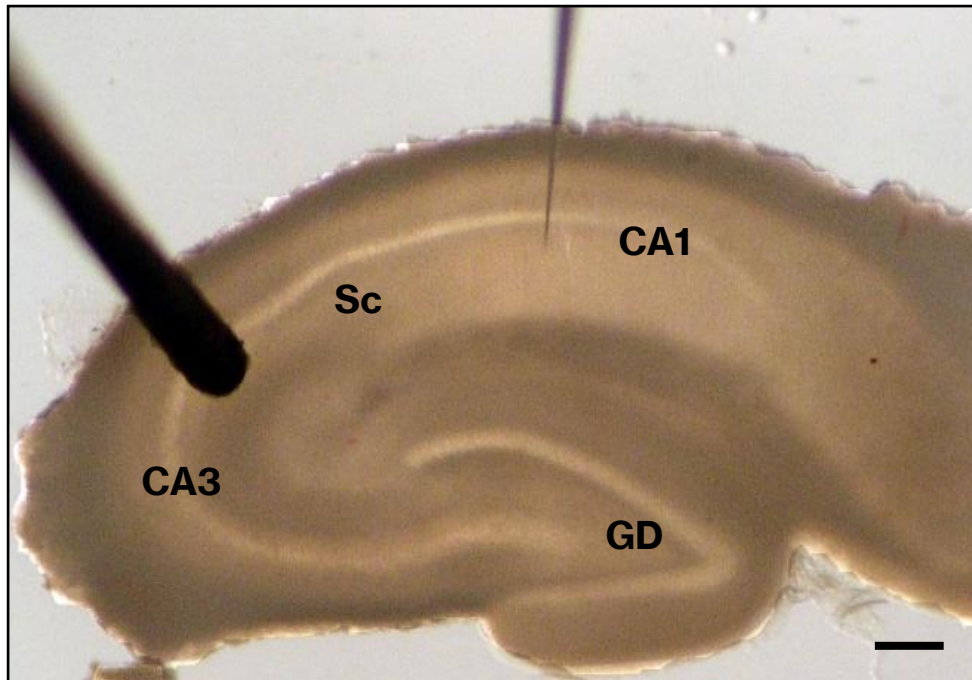
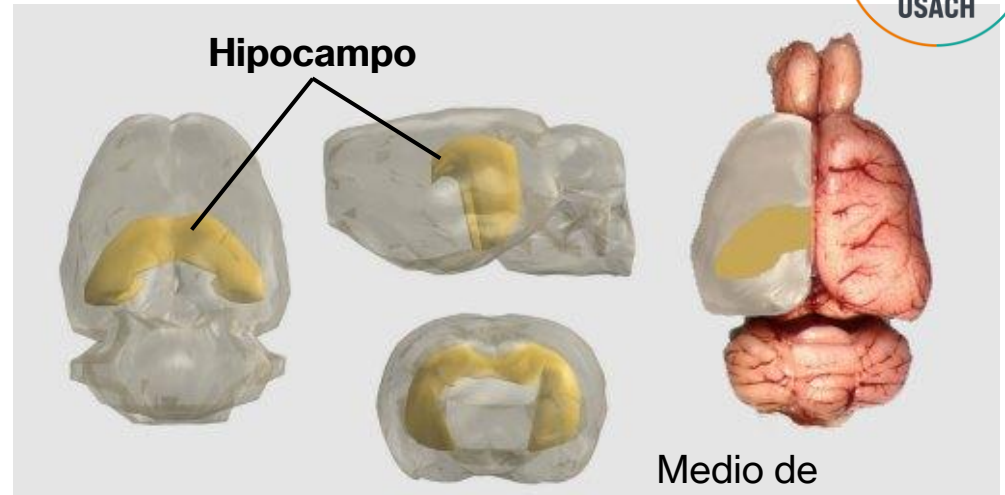
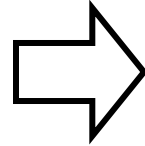


IN VITRO RECORDING TECHNIQUES

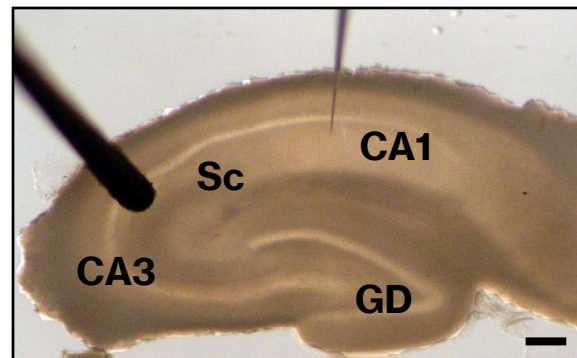
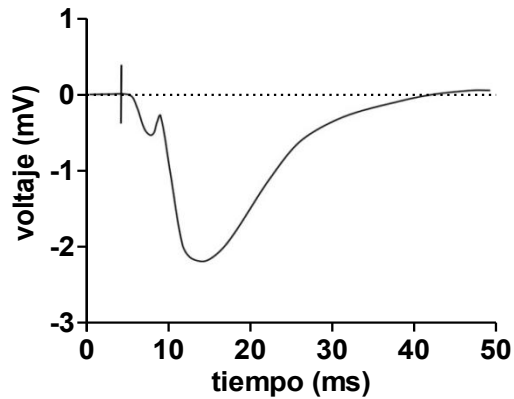
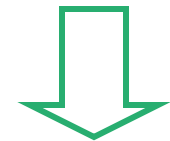
Recording from brain slices



IN VITRO RECORDING TECHNIQUES



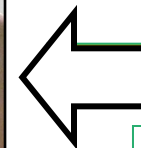
Medio de
Disección



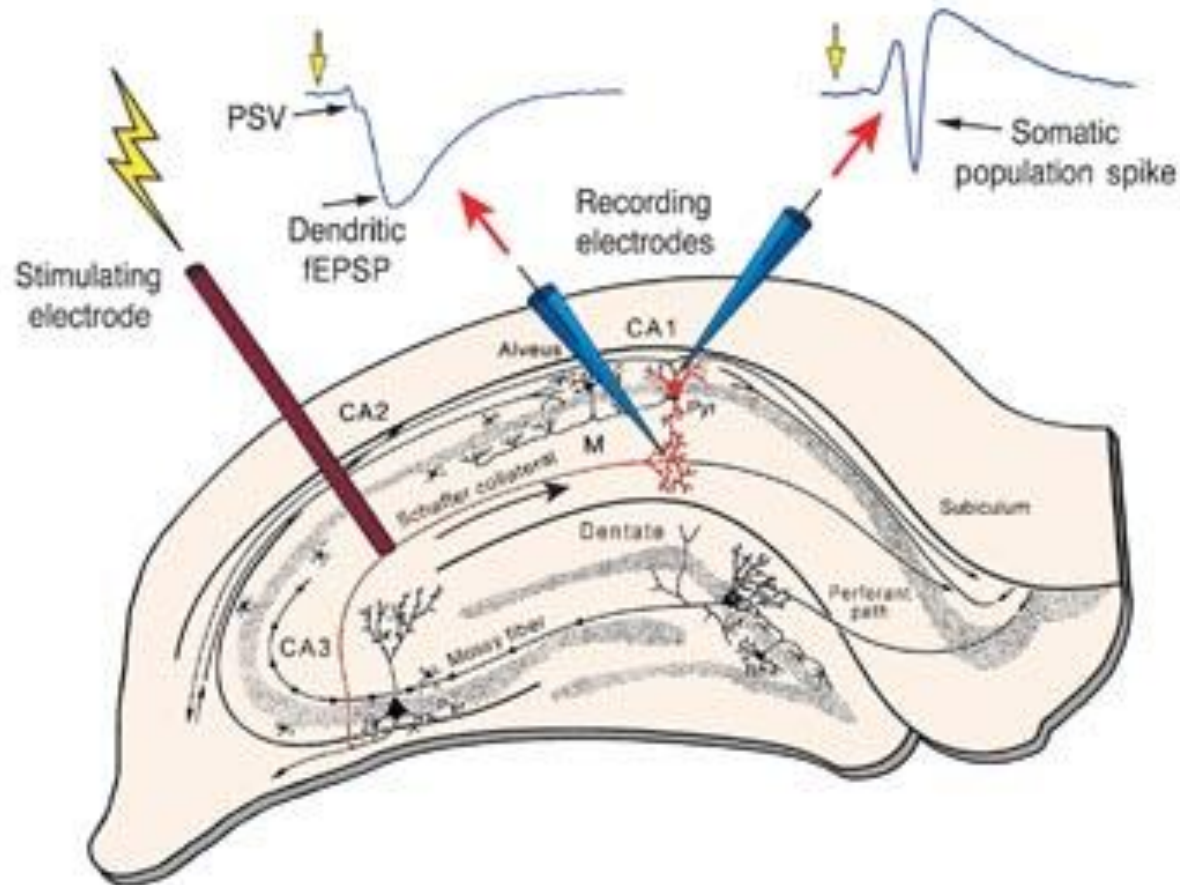
300-400 μm



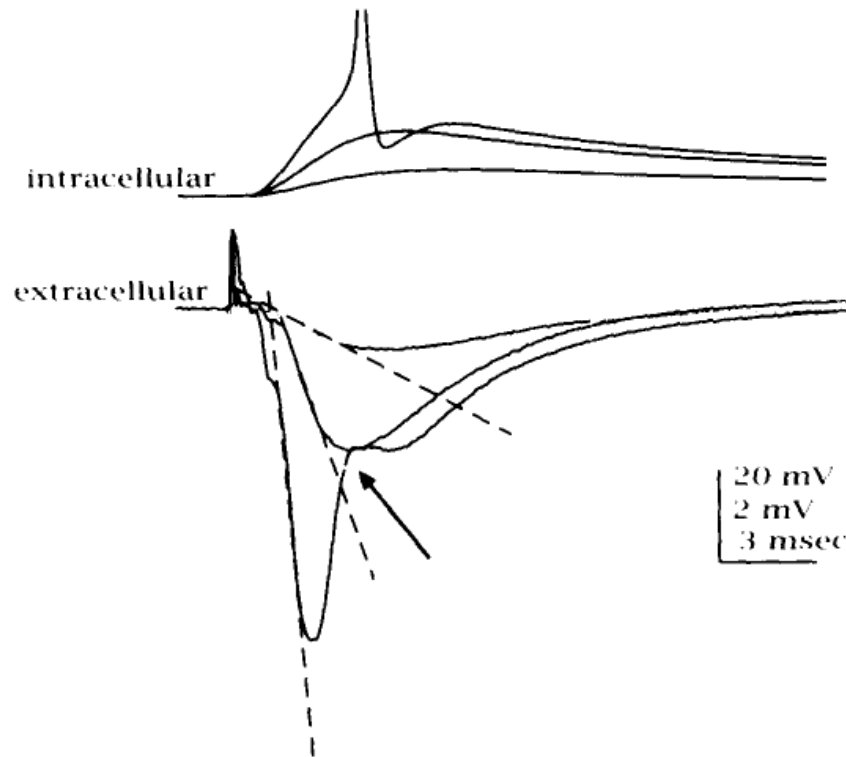
ACSF, 95% O_2 /5% CO_2



IN VITRO RECORDING TECHNIQUES



IN VITRO RECORDING TECHNIQUES



IN VITRO RECORDING TECHNIQUES

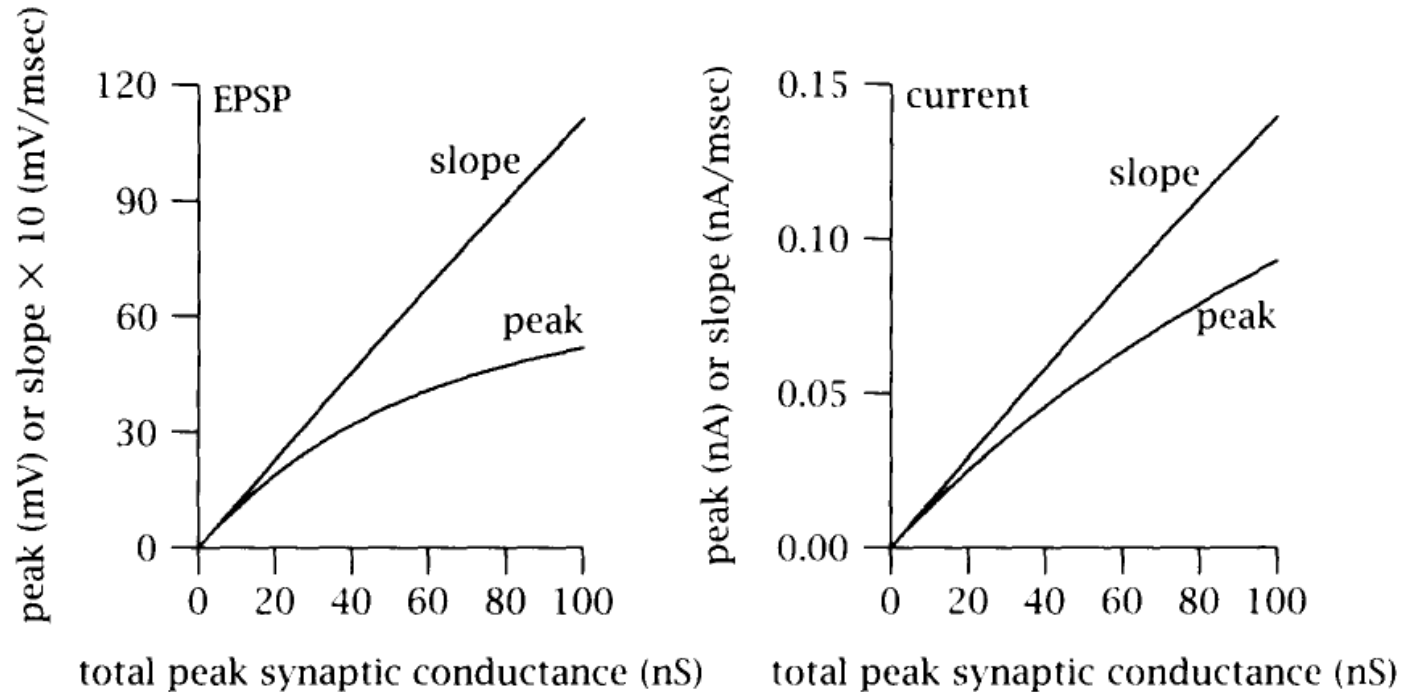


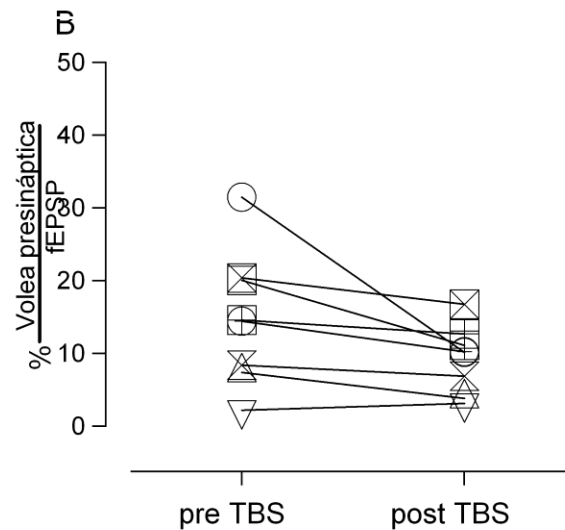
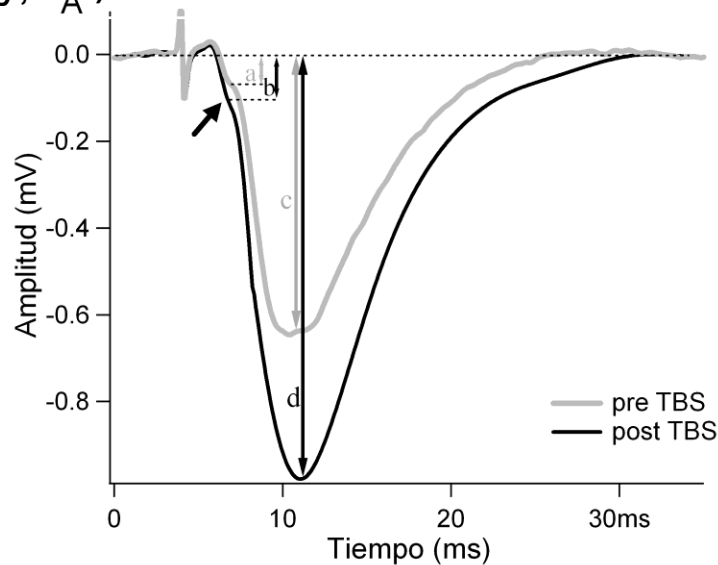
Figure 14.10 Plot of peak and slope of EPSP (left) and plot of peak and slope of EPSC (right) vs. synaptic conductance. Note that the slope is almost linearly related to the synaptic conductance whereas the peak voltage and current associated with the EPSP show pronounced nonlinear summation. (From simulations by Erik Cook.)

What else can tell us the fEPSP?

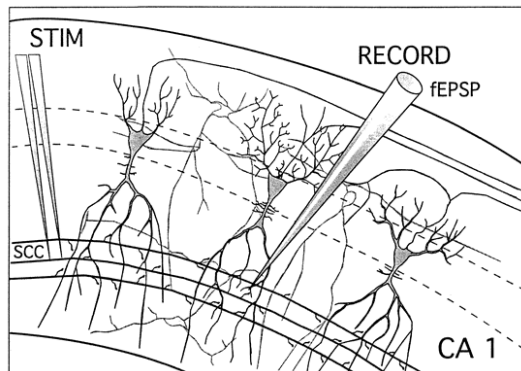


Presynaptic Volley

Following stimulation of the SC, an extracellular recording electrode in the stratum radiatum containing synapses, would record a small negative potential that results from the action potentials generated in the presynaptic fibers (Fiber Volley, FV)



Rozas C, 2010

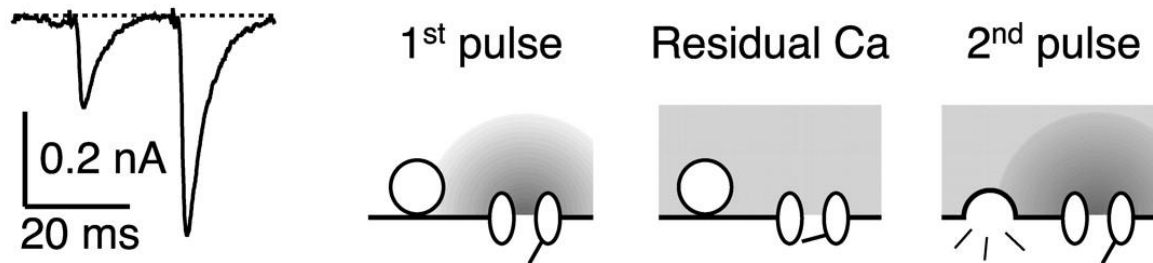


What else can tell us the fEPSP?

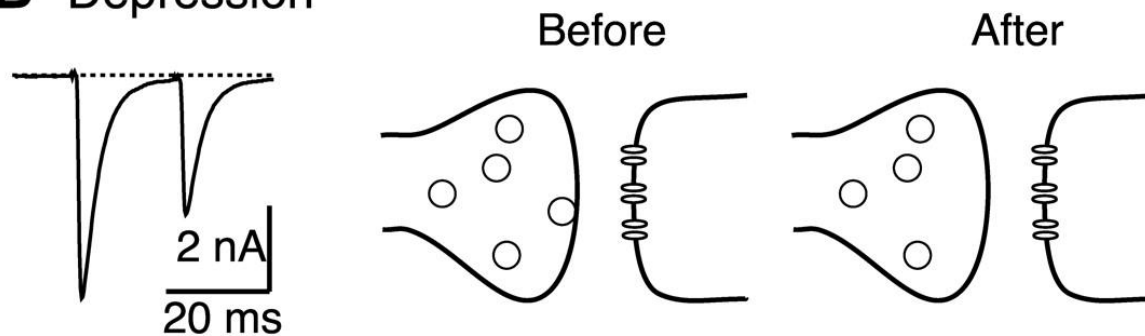


Pair pulse facilitation

A Facilitation



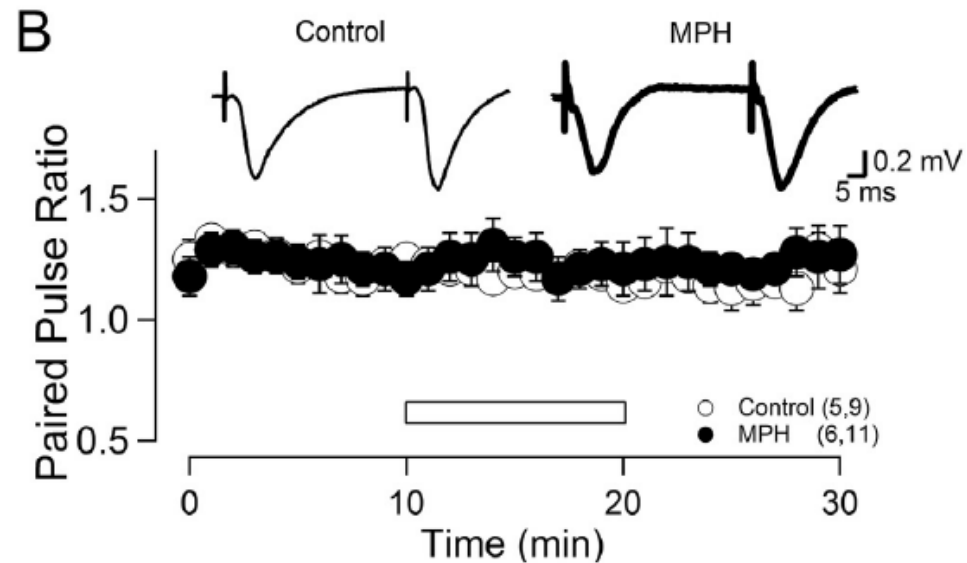
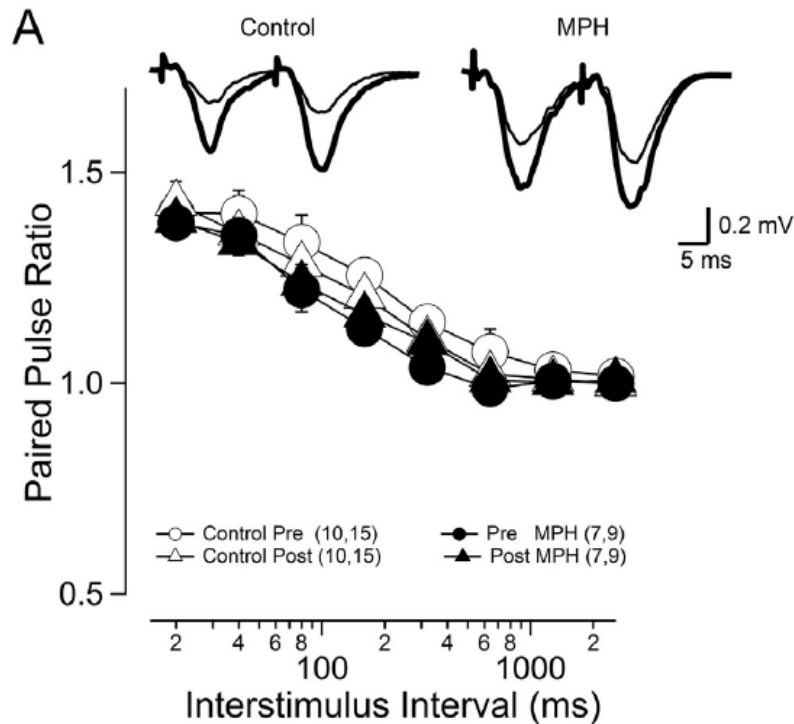
B Depression



What else can tell us the fEPSP?



Pair pulse facilitation in hippocampus



Rozas, C. et al, Neuropharmacology, 2015